

F-statistics of 1,275.76 in Model (1) and 1,220.48 in Model (2) dwarf the Stock–Yogo (2005) 10% critical value of 16.38, confirming the instrument’s strength in both specifications. Robust standard errors clustered at the reviewer invitee level are shown in parentheses. (***) $p < 0.01$, (**) $p < 0.05$, (*) $p < 0.1$).

S5 Supplementary Results for Alternative Outcome Variables

For every reviewer invitee, we summarized annual citing behavior along six dimensions. Direct Attention captures the proportion of that year’s papers in which the reviewer invitees cite the invited article itself. Breadth records the proportion that cite any work by the manuscript’s corresponding author. When those citations occur, we measure their intensity: Depth is the mean count of such citations per reviewer invitee’s paper, and Normalized Depth rescales the same counts by each paper’s reference-list length before averaging. To characterize variety, we calculate Diversity, the number of distinct corresponding author papers cited in that year. Finally, we quantify Prominence by identifying the earliest position at which the author’s work appears in a paper’s reference list, taking the reciprocal of this position as the measure. For cases where no citation occurs, we assign a value of zero. Earlier positions imply greater importance or centrality to the paper. All six metrics are computed for every calendar year from ten years before to five years after each invitation, using publication and reference data drawn from SciSciNet and WoS described above.

Table S5 presents an overview of the outcome measures employed in the analysis, detailing their conceptual motivation and formal definitions.

TABLE S5: SUMMARY OF OUTCOME VARIABLES

Outcome Variable	Explanation	Definition
Direct Attention	Measures whether researchers later cite the specific manuscript they were invited to review, conditional on that manuscript being published. Indicates direct influence from the exposure.	Proportion of reviewer invitees' publications each year that cite the invited manuscript: $\frac{N_{pubs_citing_manuscript}}{N_{pubs_per_year}}$
Breadth	Assesses the extent to which reviewer invitees cite any work by the corresponding author (beyond the focal manuscript), capturing broader engagement with the author's research.	Proportion of reviewer invitees' publications each year that cite any paper by the corresponding author: $\frac{N_{pub_citing_author}}{N_{pubs_per_year}}$
Depth (Figure 2)	Evaluates how extensively a reviewer invitee cites the corresponding author's work in each publication. Higher values reflect more intensive engagement.	Average number of references to the corresponding author each year: $\frac{1}{N_{pubs}} \sum_{i=1}^{N_{pubs}} \#refs_i^{to_author}$
Normalized Depth	Adjusts depth by accounting for total number of references in each publication. Captures relative intensity of citations to the author.	Average proportion of references per publication attributed to the corresponding author each year: $\frac{1}{N_{pubs}} \sum_{i=1}^{N_{pubs}} \left(\frac{\#refs_i^{to_author}}{\#refs_i} \right)$
Diversity	Captures the range of the corresponding author's work cited by the reviewer invitees. Higher values indicate engagement with multiple distinct papers, not just repeated citation of one.	Total number of unique papers by the corresponding author cited each year: $N_{unique_cited_papers_by_author}$
Prominence	Indicates the prominence of the corresponding author's work in the reviewer invitee's reference list, based on citation position. Earlier positions imply greater importance or centrality to the paper.	The reciprocal of earliest reference list position of the author's work across reviewer invitees' publications each year: $\frac{1}{\sum_{i=1}^{N_{pubs}} \min(Ref_Position_{author,i})}$

Table S5. Definitions of outcome variables capturing downstream citation behavior. The table lists each outcome variable's conceptual focus—ranging from direct attention to the invited manuscript to breadth, depth, diversity, and prominence of citations to the corresponding author—and provides its exact operational formula.

We compare reviewer invitees who click-declined versus those who auto-declined the invitation across six citation-based metrics mentioned above using two-sample t-tests for both post-invitation (Figure 4) and pre-invitation (Figure S3) periods. For pre-invitation, the two groups do not differ significantly in direct attention, diversity, or prominence (Figure S3AEF), but exhibit significant difference in breadth (Figure S3B, $p = 0.001$) and depth (Figure S3C, $p = 0.001$), and a marginal difference in normalized depth (Figure S3D, $p = 0.077$). Although some dimensions show statistically significant pre-invitation differences, they consistently run in the opposite direction of our primary post-invitation findings. This pattern suggests that our observed post-invitation differences likely represent conservative estimates of the peer review exposure effect.

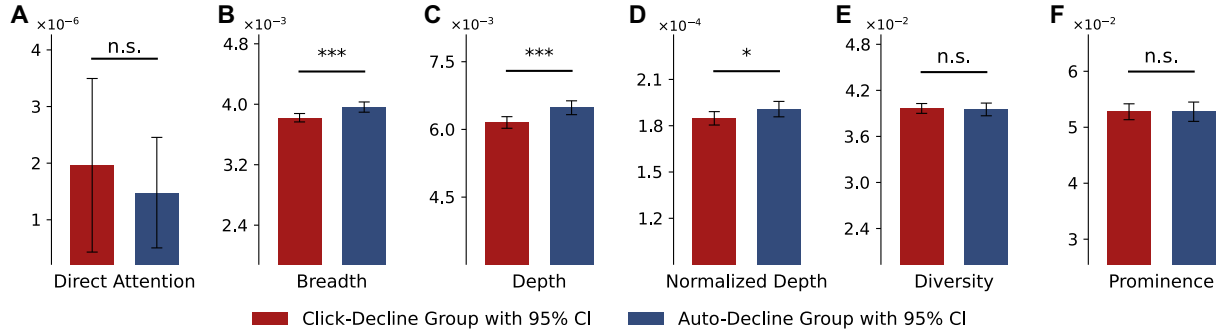


Figure S3. Pre-invitation comparisons across six dimensions. We compare the Click-Denied and Auto-Denied groups using two-sample t-tests on six citation-based outcome measures, for pre-invitation periods. Group means are displayed with 95% confidence intervals, with significance levels and conceptual groupings annotated. Outcome variables include: (A) Direct Attention; (B) Breadth; (C) Depth; (D) Normalized Depth; (E) Diversity; and (F) Prominence. Color coding indicates conceptual clusters. Pre-invitation differences between Click-Denied and Auto-Denied groups were examined across the six dimensions of citation behavior. While several dimensions show statistically significant pre-invitation difference (Breadth, $p = 0.001$; Depth, $p = 0.001$; and Normalized Depth, $p = 0.077$), these differences run notably in directions opposite to our primary findings. This suggests our post-invitation differences represent conservative estimates of the peer review effect. (***) $p < 0.01$, (**) $p < 0.05$, (*) $p < 0.1$).

The DiD specification in Table S6 is identical to the baseline model reported in the main text, with the only difference being the dependent variable:

$$y_{it} = \beta_0 \text{Treat}_i \times \text{Post}_t + \beta_1 \text{Treat}_i + \beta_2 \text{Post}_t + \beta_3 C_i + \epsilon_{it}$$

Across all six measures—Direct Attention, Breadth, Depth, Normalized Depth, Diversity, and Prominence—the point estimates are positive and statistically significant ($p < 0.01$). The

results indicate that exposure via the review invitation not only boosts the likelihood of citing the specific manuscript but also broadens, deepens, diversifies, and elevates the author's presence in reviewer invitees' subsequent research.

TABLE S6: ALTERNATIVE OUTCOME VARIABLE ESTIMATES

	Direct Attention	Breadth	Depth	Normalized Depth	Diversity	Prominence
$Treat_i \times Post_t$	0.0001*** (0.0000)	0.0012*** (0.0002)	0.0021*** (0.0004)	0.0001*** (0.0000)	0.0259*** (0.0028)	0.0377*** (0.0056)
$Treat_i$	0.0000 (0.0000)	-0.0001* (0.0001)	-0.0003* (0.0002)	0.0000 (0.0000)	0.0001 (0.0012)	-0.0000 (0.0017)
$Post_t$	0.0004*** (0.0000)	0.0050*** (0.0001)	0.0087*** (0.0003)	0.0002*** (0.0000)	0.0726*** (0.0019)	0.1352*** (0.0040)
Const	0.0000*** (0.0000)	0.0040*** (0.0001)	0.0065*** (0.0001)	0.0002*** (0.0000)	0.0395*** (0.0010)	0.0523*** (0.0013)
Observations	4,755,571	4,755,571	4,755,571	4,755,571	4,755,571	4,755,571
No. clusters	450,737	450,737	450,737	450,737	450,737	450,737
R-squared	0.0009	0.0026	0.0014	0.0010	0.0035	0.0019

Table S6. The effect of exposure via review invitation on six dimensions. The table reports the Difference-in-Differences estimates across six dimensions. We run DiD regressions restricted to Click-Delay and Auto-Delay on six citation-based outcome measures. The treatment effect is positive and highly significant for every outcome variable: Direct Attention = 0.0001 ($p = 0.007$), Breadth = 0.0012 ($p = 0.000$), Depth = 0.0021 ($p = 0.000$), Normalized Depth = 0.0001 ($p = 0.000$), Diversity = 0.0259 ($p = 0.000$), and Prominence = 0.0377 ($p = 0.000$). These coefficients correspond to relative increases of 25%, 13%, 14%, 25%, 23% and 20% in Direct Attention, Breadth, Depth, Normalized Depth, Diversity, and Prominence, respectively, compared with the Auto-Delay group's post-invitation citation rates. Robust standard errors clustered at the reviewer invitee level are shown in parentheses. (***) $p < 0.01$, (**) $p < 0.05$, (*) $p < 0.1$).